

## EXERCISE WEEK 8

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```
1 library(dplyr)
2
3 students <- read.csv("C:/Users/thurg/Downloads/student_data.csv")
4
5 #Show the details of students whose final_exam_mark is greater than 80%.
6 students[students$final_exam_mark > 80, ]
7
8 #Arrange the dataset in ascending order based on course_work_mark.
9 mydata1 <- students %>%
10   arrange(coursework_mark)
11
12 View(mydata1)
13
14 #Display only the student_name and final_exam_mark attributes.
15 students[, c("student_name", "final_exam_mark")]
16
17 #Specify the number of rows and columns.
18 nrow(students)
19 ncol(students)
20
21 #Obtain the dataset details.
22 str(students)
23 summary(students)
24 head(students)
25 tail(students)
26
27 #Detect outliers using a boxplot.
28 Q1 <- quantile(students$final_exam_mark, 0.25)
29 Q3 <- quantile(students$final_exam_mark, 0.75)
30
31 IQR_value <- IQR(students$final_exam_mark)
32
33 lower_limit <- Q1 - 1.5 * IQR_value
34 upper_limit <- Q3 + 1.5 * IQR_value
35
36 students[students$final_exam_mark < lower_limit |
37   students$final_exam_mark > upper_limit, ]
38
39
```

```
Console Terminal Background Jobs
R 4.6.0 ~ /

> library(dplyr)
>
> students <- read.csv("C:/Users/thurg/Downloads/student_data.csv")
>
> #Show the details of students whose final_exam_mark is greater than 80%.
> students[students$final_exam_mark > 80, ]
  student_id student_name gender coursework_mark final_exam_mark
7      S007   David Jones   Male              64                84
32     S032  Joseph Smith   Male              94                82
44     S044 Jessica Brown Female              77                84
>
> #Arrange the dataset in ascending order based on course_work_mark.
> mydata1 <- students %>%
+   arrange(coursework_mark)
>
> View(mydata1)
>
> #Display only the student_name and final_exam_mark attributes.
> students[, c("student_name", "final_exam_mark")]
  student_name final_exam_mark
1 Elizabeth Miller              41
2 William Garcia              63
3 Sarah Johnson              64
4 William Davis              62
5 John Smith              56
6 Sarah Rodriguez              35
7 David Jones              84
8 Thomas Martinez              60
9 Michael Garcia              64
10 Jessica Garcia              65
11 David Miller              49
12 Mary Garcia              55
13 John Smith              53
14 John Brown              58
15 Elizabeth Garcia              51
16 Robert Rodriguez              55
17 Linda Garcia              74
18 Sarah Rodriguez              70
19 Susan Williams              44
20 Joseph Rodriguez              62
21 Robert Miller              62
```

```
Console Terminal Background Jobs
R 4.6.0 ~ /
20 Joseph Rodriguez 62
21 Robert Miller 62
22 Robert Miller 54
23 David Martinez 55
24 William Jones 64
25 Joseph Williams 73
26 Charles Johnson 57
27 Charles Johnson 48
28 Charles Garcia 39
29 Linda Smith 68
30 Barbara Williams 55
31 Joseph Williams 66
32 Joseph Smith 82
33 Richard Miller 42
34 Richard Martinez 67
35 Barbara Miller 54
36 Karen Jones 45
37 David Davis 38
38 Elizabeth Miller 34
39 Linda Williams 70
40 Sarah Miller 51
41 Barbara Martinez 55
42 Sarah Miller 73
43 Patricia Williams 38
44 Jessica Brown 84
45 Susan Brown 77
46 Mary Davis 60
47 Linda Rodriguez 54
48 Jessica Garcia 54
49 Linda Davis 53
50 Robert Brown 57
>
> #Specify the number of rows and columns.
> nrow(students)
[1] 50
> ncol(students)
[1] 5
>
> #Obtain the dataset details.
> str(students)
'data.frame': 50 obs. of 5 variables:
```

```

>
> #Obtain the dataset details.
> str(students)
'data.frame': 50 obs. of 5 variables:
 $ student_id : chr "S001" "S002" "S003" "S004" ...
 $ student_name : chr "Elizabeth Miller" "William Garcia" "Sarah Johnson" "William Davis" ...
 $ gender : chr "Female" "Male" "Female" "Male" ...
 $ coursework_mark: int 40 59 79 56 74 41 64 73 70 67 ...
 $ final_exam_mark: int 41 63 64 62 56 35 84 60 64 65 ...
> summary(students)
 student_id student_name gender coursework_mark final_exam_mark
Length :50 Length :50 Length :50 Min. :34.00 Min. :34.00
N.unique :50 N.unique :42 N.unique : 2 1st Qu.:51.00 1st Qu.:51.50
N.blank : 0 N.blank : 0 N.blank : 0 Median :62.00 Median :56.50
Min.nchar: 4 Min.nchar:10 Min.nchar: 4 Mean :62.42 Mean :57.88
Max.nchar: 4 Max.nchar:17 Max.nchar: 6 3rd Qu.:73.00 3rd Qu.:64.75
Max. :97.00 Max. :84.00

> head(students)
 student_id student_name gender coursework_mark final_exam_mark
1 S001 Elizabeth Miller Female 40 41
2 S002 William Garcia Male 59 63
3 S003 Sarah Johnson Female 79 64
4 S004 William Davis Male 56 62
5 S005 John Smith Male 74 56
6 S006 Sarah Rodriguez Female 41 35
> tail(students)
 student_id student_name gender coursework_mark final_exam_mark
45 S045 Susan Brown Female 77 77
46 S046 Mary Davis Female 70 60
47 S047 Linda Rodriguez Female 50 54
48 S048 Jessica Garcia Female 63 54
49 S049 Linda Davis Female 61 53
50 S050 Robert Brown Male 73 57
>
> #Detect outliers using a boxplot.
> Q1 <- quantile(students$final_exam_mark, 0.25)
> Q3 <- quantile(students$final_exam_mark, 0.75)
>
> IQR_value <- IQR(students$final_exam_mark)
>
>
> #Detect outliers using a boxplot.
> Q1 <- quantile(students$final_exam_mark, 0.25)
> Q3 <- quantile(students$final_exam_mark, 0.75)
>
> IQR_value <- IQR(students$final_exam_mark)
>
> lower_limit <- Q1 - 1.5 * IQR_value
> upper_limit <- Q3 + 1.5 * IQR_value
>
> students[students$final_exam_mark < lower_limit |
+ students$final_exam_mark > upper_limit, ]
[1] student_id student_name gender coursework_mark final_exam_mark
<0 rows> (or 0-length row.names)
>

```

